

Clinical: Disorders of Fluid Balance and Hyponatremia

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Session Objectives

- Explain water balance within the body.
- Illustrate the pathophysiology of hypernatremia and hypovolemia.
- Differentiate the causes of hypernatremia and hypovolemia.
- Illustrate how to diagnose and treat hypernatremia.
- Explain diabetes insipidus.

Question

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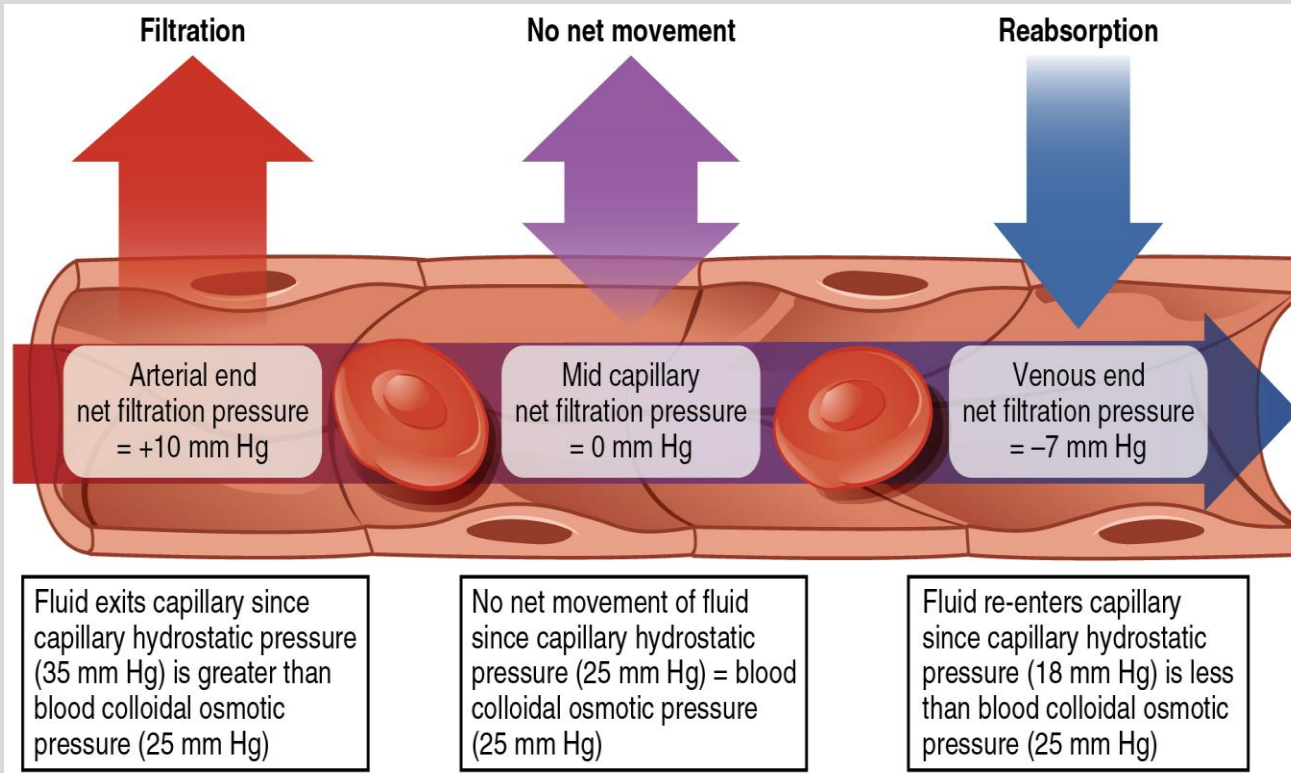
**How much of the total body water
is located within the intracellular fluid?**

- A. 10%
- B. 25%
- C. 33%
- D. 67%

Question

Excretion or retention of electrolyte-free water by the kidney is modulated through which of the following hormones?

- A. T3
- B. Oxytocin
- C. Epinephrine
- D. ADH



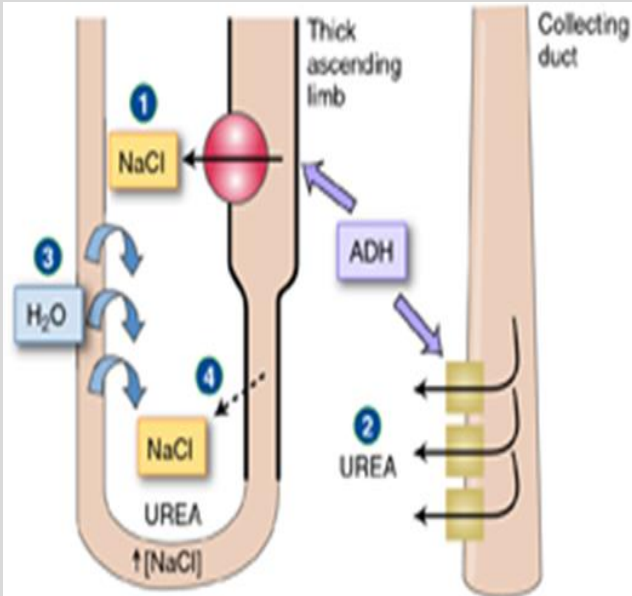
Water Balance

- Arginine vasopressin (AVP) = Vasopressin
=Antidiuretic Hormone (ADH)
- Synthesized in magnocellular neurons in hypothalamus
- Released from posterior pituitary gland

ADH

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Source: Jonathan D. Kibble, Colby R. Halsey: *The Big Picture Medical Physiology*:
www.accessphysiotherapy.com

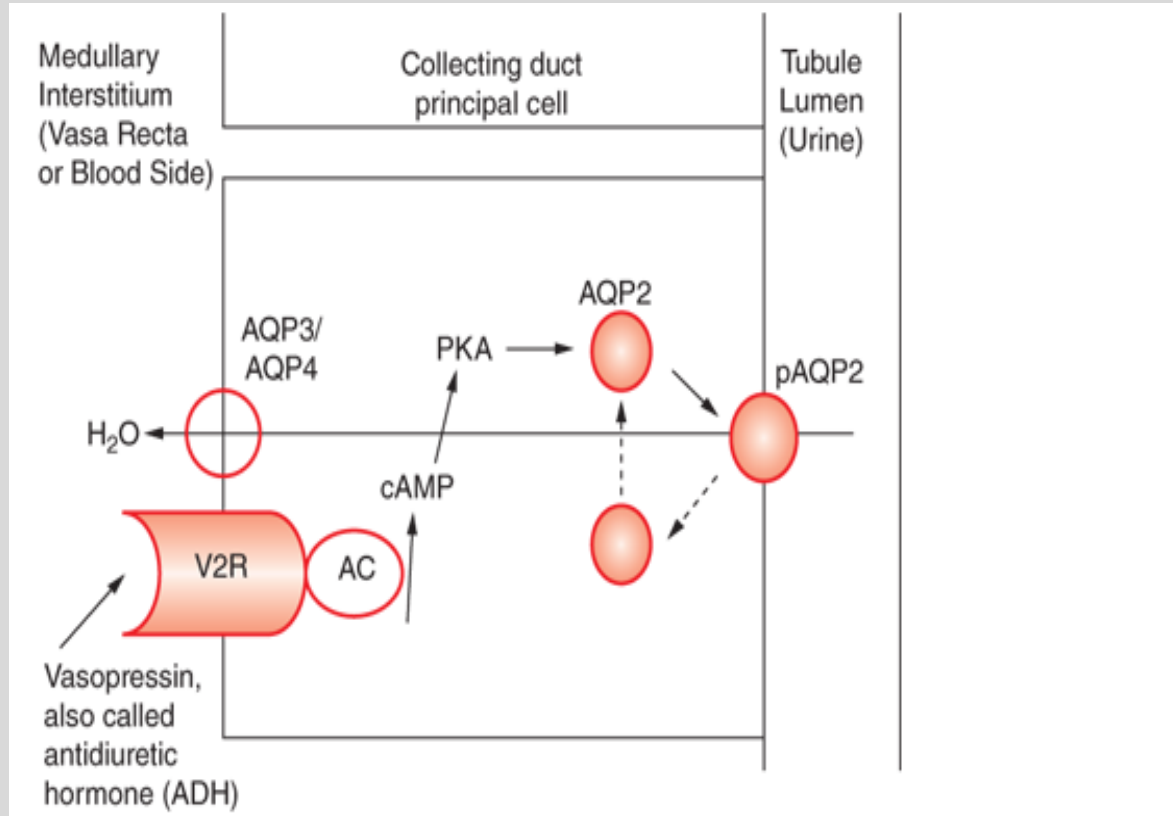
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Source: Renal Physiology and Acid-Base Balance, Kibble JD, Halsey CR. *Medical Physiology: The Big Picture*; 2015. Available at: <https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=75577187&gbosContainerID=0&gbosid=0&groupID=0§ionId=75577039&multimediaId=undefined> Accessed: July 06, 2020

ADH

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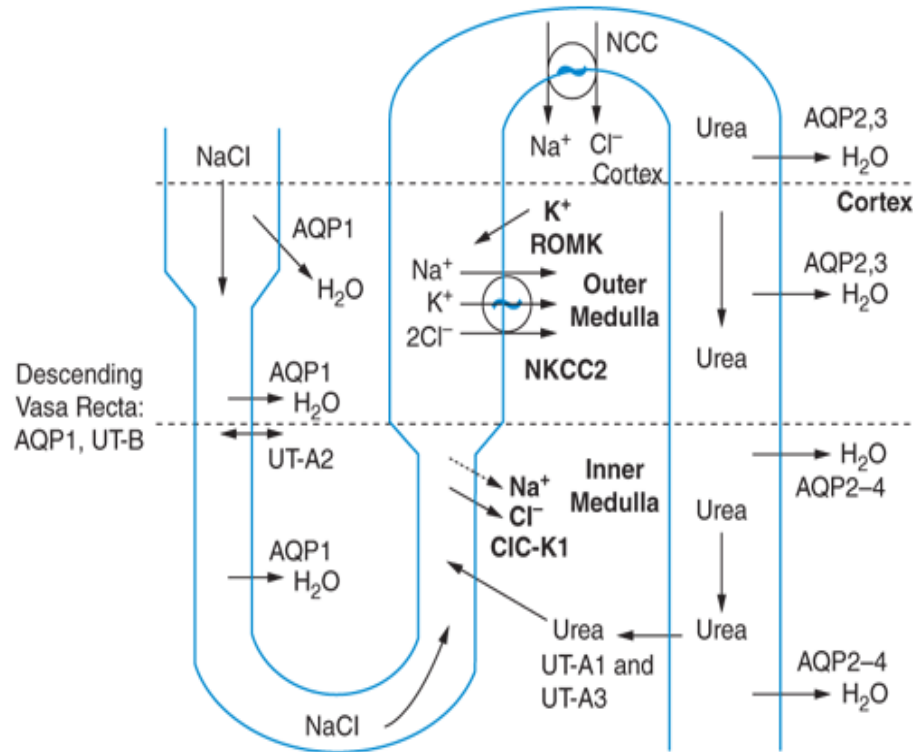
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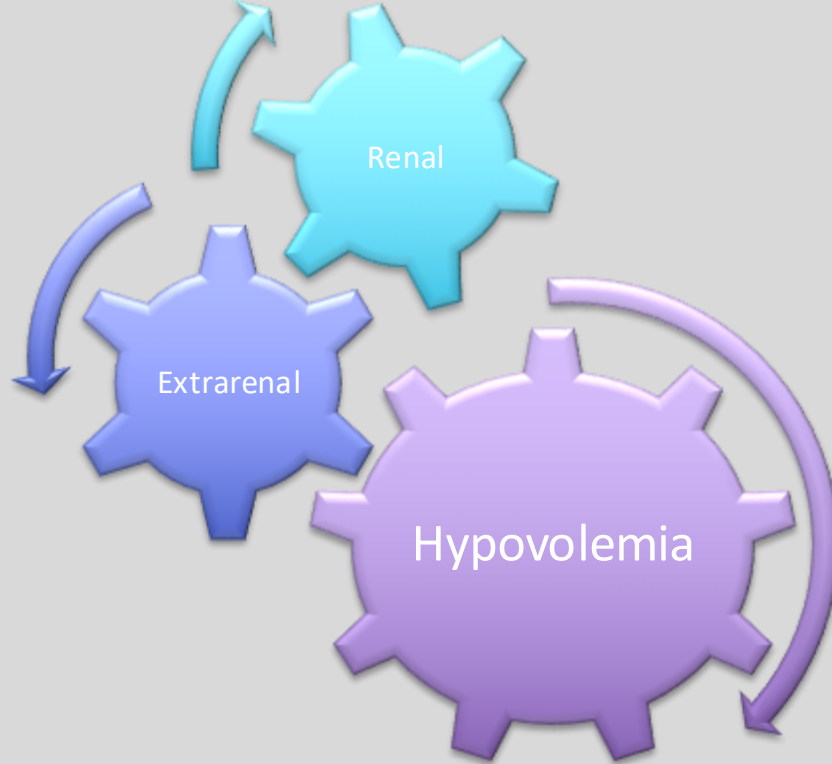
ADH

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Hypovolemia causes



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Clinical Scenario

- Mr. R is an 80-year-old nursing home resident with a history of severe dementia. He has been brought to the emergency department with lethargy and confusion. Serum chemistries reveal a sodium level of 168 mEq/L.

Hypernatremia

- Plasma Sodium concentration > 145 mmol/L
- Hypernatremia is a manifestation of disease or disorder in body fluid maintenance.
- Less common than hyponatremia
- Mortality rates as high as 40-60%

Hypernatremia DDx

- The differential diagnosis of hypernatremia is simpler than hyponatremia.
- 3 main situations:
 - (1) impaired water intake
 - (2) hyperglycemic hyperosmolar state
 - (3) diabetes insipidus

Hypernatremia Physiology

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- Hypernatremia and hyperosmolality potent stimulators of ADH & thirst
- Promotes drinking and hypernatremia
- Hypernatremia occurs
 - Unaware of thirst
 - Unable to obtain water

Source: Stern SC. Hyponatremia and Hypernatremia. In: Stern SC, Cifu AS, Altkorn D. eds. *Symptom to Diagnosis: An Evidence-Based Guide*, 4e. McGraw-Hill

Common Clinical Scenarios

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- Infants
- Debilitated elderly patients with severe dementia
- 50% of elderly patients, a superimposed process (ie, pneumonia, urinary tract infection, or cerebrovascular accident)
- The 30-day mortality in elderly hypernatremic patients has been reported at 41.5%.

Source: Stern SC. Hyponatremia and Hypematremia. In: Stern SC, Cifu AS, Altkorn D. eds. *Symptom to Diagnosis: An Evidence-Based Guide*, 4e. McGraw-Hill

Pathophysiology

- Normal insensible water losses or increased water loss (ie, from diarrhea)
- Not matched by oral intake and hypernatremia develops.
- Kidneys maximize water reabsorption

Pathophysiology

- Hypernatremia with marked hyperglycemia.
- Osmotic diuresis results in a free water loss
 - w/ altered sensorium less water intake may result in hypernatremia
- Magnitude may be hidden in initial labs
- Hyperglycemia draws H₂O from ICF to ECF diluting the sodium concentration.

Clinical Findings

- Dehydration
 - Tachycardia
 - Orthostatic hypotension
 - Dry mucous membranes
- Hypernatremia
 - depressed sensorium
 - focal deficits
 - seizures

Finding	Sensitivity	Specificity
Tachycardia	17.8%	94%
Orthostatic Hypotension	61.5%	50.6%
Decreased skin turgor	73.3%	79%
Dry mucosa	49%	87.8%

Differential Diagnosis

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Diagnostic Hypotheses	Demographics, Risk Factors, Symptoms and Signs	Important Tests
Leading Hypothesis		
Inadequate water consumption	Elderly patients, history of neurologic or physical disability Concomitant illness Hypotension, tachycardia	Urine osmolality > 600 mOsm/L Chest radiograph Urinalysis and culture Electrolytes, BUN, creatinine, glucose
Active Alternative—Most Common and Must Not Miss		
Hyperglycemia	Diabetes mellitus, concurrent illness, hypotension, tachycardia	Markedly elevated serum glucose
Active Alternative		
Diabetes insipidus	Complaints of polydipsia, polyuria	Urine osmolality < 600 mOsm/L
Central diabetes insipidus	History of CNS trauma, surgery, CVA, sarcoidosis	ADH levels low Administration of exogenous ADH markedly increases urine osmolality
Nephrogenic diabetes insipidus	Lithium ingestion	ADH levels elevated Exogenous ADH minimally elevates urine osmolality

Source: Stern SC. Hyponatremia and Hypermnatremia. In: Stern SC, Cifu AS, Altkorn D. eds. *Symptom to Diagnosis: An Evidence-Based Guide*, 4e. McGraw-Hill

Clinical Scenario

- Mr. R is minimally responsive to stimuli.
- Vital signs are BP, 110/70 mm Hg; pulse, 110 bpm; temperature, 100.6°F; RR, 20 breaths per minute.
- His oral mucosa is parched and his axilla dry.
- Lung exam is difficult to evaluate due to poor effort.
- Cardiac exam reveals tachycardia; neck veins are flat. There is no S₃ or S₄.

Clinical Scenario

- Laboratory findings:
- Na, 168 mEq/L (135 and 145 mEq/L)
- K, 4.2 mEq/L (3.6 to 5.2 mEq/L)
- HCO_3^- , 24 mEq/L (23 to 29 mEq/L)
- Chloride, 134 mEq/L (97-107 mEq/L)
- BUN, 45 mg/dL (7-20 mg/dL)
- Creatinine, 1 mg/dL (0.84 to 1.21 mg/dL)
- Serum glucose is 110 mg/dL (70 and 99 mg/dL)

Clinical Scenario

- **Leading Hypothesis: Hypernatremia Secondary to Inadequate Water Intake**
- Hypernatremia due to inadequate water ingestion
 - Result of altered neurologic status or physical disability
- Superimposed illness may worsen cognitive function, decrease oral intake, and promote hypernatremia.
- **Evidenced based diagnosis**
- Confirmed by hypernatremia, increased urine osmolality, and absence of hyperglycemia.

Clinical Scenario

- Making Diagnosis
- Mr. R's urine osmolality is 950 mOsm/L (500-850 mOsm/kg)
- A chest film reveals a right lower lobe pneumonia. Blood cultures grow *Streptococcus pneumoniae*.

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Treatment

- Identify the underlying cause and attempt to correct it
- Correct the water deficit

Water Deficit

$$([\text{Serum Na} / 140] - 1) \times (\text{weight kg} \times 0.6)$$

Example: Middle-aged 60 kg woman with a serum sodium concentration of 168 meq/L, total body water is about 50 percent of body weight, and the water deficit can be approximated from:

$$\text{Water deficit} = (168/140 - 1) \times (0.5 \times 60) = 6 \text{ liters}$$

$$\text{Water deficit} = (1.2 - 1) \times (30)$$

$$\text{Water deficit} = (.2) \times (30) = 6 \text{ L}$$

Water Deficit

Change in serum Na^+

$$= (\text{infusate } \text{Na}^+ - \text{serum } \text{Na}^+) \div (\text{TBW} + 1)$$

- 5% dextrose in water ($\text{D}_5 \text{W}$): 0 mmol/L
- 0.9% sodium chloride in water (0.9NS): 154 mmol/L

Example

Change in serum $\text{Na} = (0 - 168) \div (30 + 1) = -5.4 \text{ mmol}$

Therefore 1 L of D5W will decrease Na by 5.4 mmol

Water Deficit

Goal is to reduce serum sodium by **4 to 6 meq/L not exceeding 10 meq/L in 24 hours.**

- $(6\text{meq/L}/5.4\text{meq}) = 1.1\text{L}$
- 1.1L plus 1-1.5 L obligatory water loss
- = 2.5 L over 24 hours (104cc/h) of D5W

Clinical Scenario

- Mr. R is given D5W. His body weight is measured at 140 lbs (63 kg).
- Calculate free water administration
- Piperacillin-tazobactam to treat his aspiration pneumonia.
- 3 days later electrolytes normal and neurologic function returns
- Discharged to nursing home w/ oral antibiotics

Clinical Scenario 2

- 79-year-old female with a history of mild dementia presents with worsening confusion over the last 2 days.
- She lives in a nursing home and has recently been treated with antibiotics for a urinary tract infection.
- You have very little information available when she presents to the ED. However, she has been healthy otherwise and is on no other medication.

Clinical Scenario 2

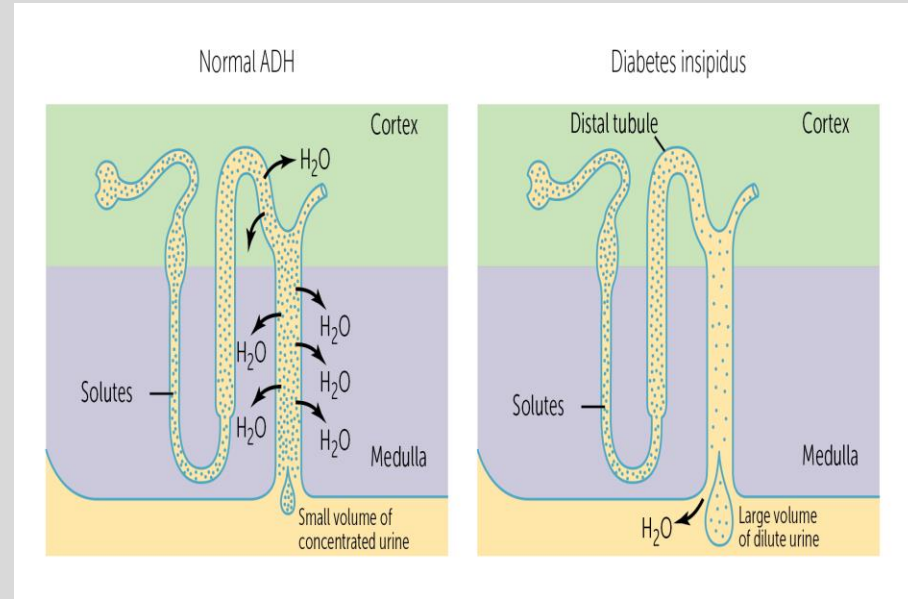
- Her temperature is 100.1° F, pulse 90, respiratory rate 18, and blood pressure 118/60 mm Hg.
- She is disoriented and lethargic.
- Examination of the heart, lungs, and abdomen is unremarkable. Poor skin turgor, dry mucous membranes.
- Increased Na, Plasma osmolality, decreased Urine osmolality

Diabetes Insipidus

- Excessive water excretion
- Decreased or resistance to ADH
- ADH causes concentrated urine increase total body water.
- High plasma sodium concentration
- Increased plasma osmolality
- Low urine osmolality

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Diabetes Insipidus

- Central Diabetes Insipidus
 - Deficient secretion of antidiuretic hormone (ADH)
- Nephrogenic Diabetes Insipidus
 - Normal ADH secretion
 - Renal resistance to water-retaining effect
 - Lithium dysfunction of the aquaporin-2 water channel (Collecting ducts)

Central Diabetes Insipidus

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Etiologies

- Idiopathic
- Head trauma
- Postoperative (hypophysectomy)
- Central nervous system tumors
- Infections
- Basilar meningitis
- Granulomatous disease
- Cerebrovascular disease

Source: Bichet, D. (2022) Diagnosis of polyuria and diabetes insipidus. *UpToDate*.

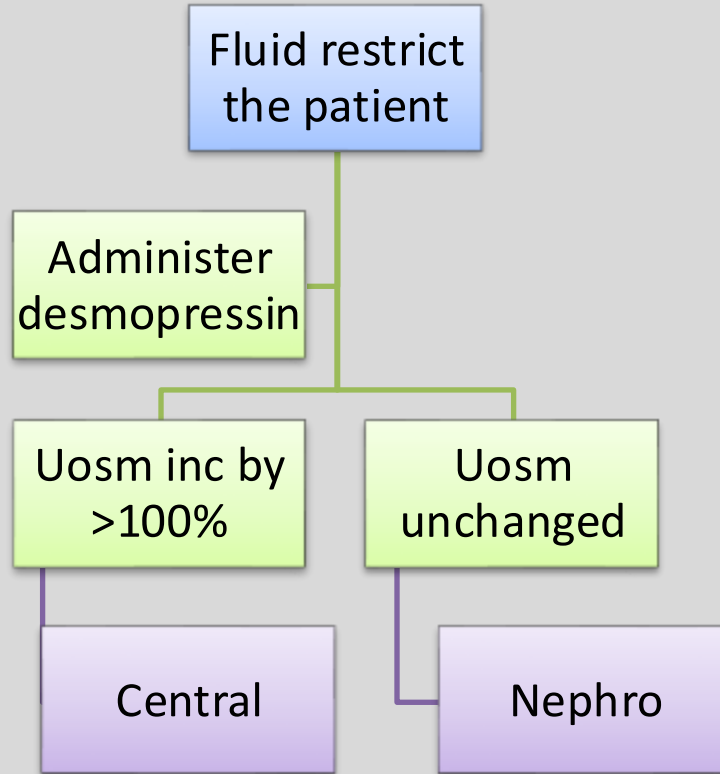
- **Etiologies**

- Congenital (Vasopressin V2-receptor defect or Aquaporin-2 water channel defect)
- Drugs (Lithium, Amphotericin B, *Cidofovir HIV Medication*)
- Multiple myeloma
- Amyloidosis
- Sarcoidosis
- Systemic lupus erythematosus
- Polycystic kidney disease
- Electrolyte disorders (Hypercalcemia and Potassium depletion)

Management

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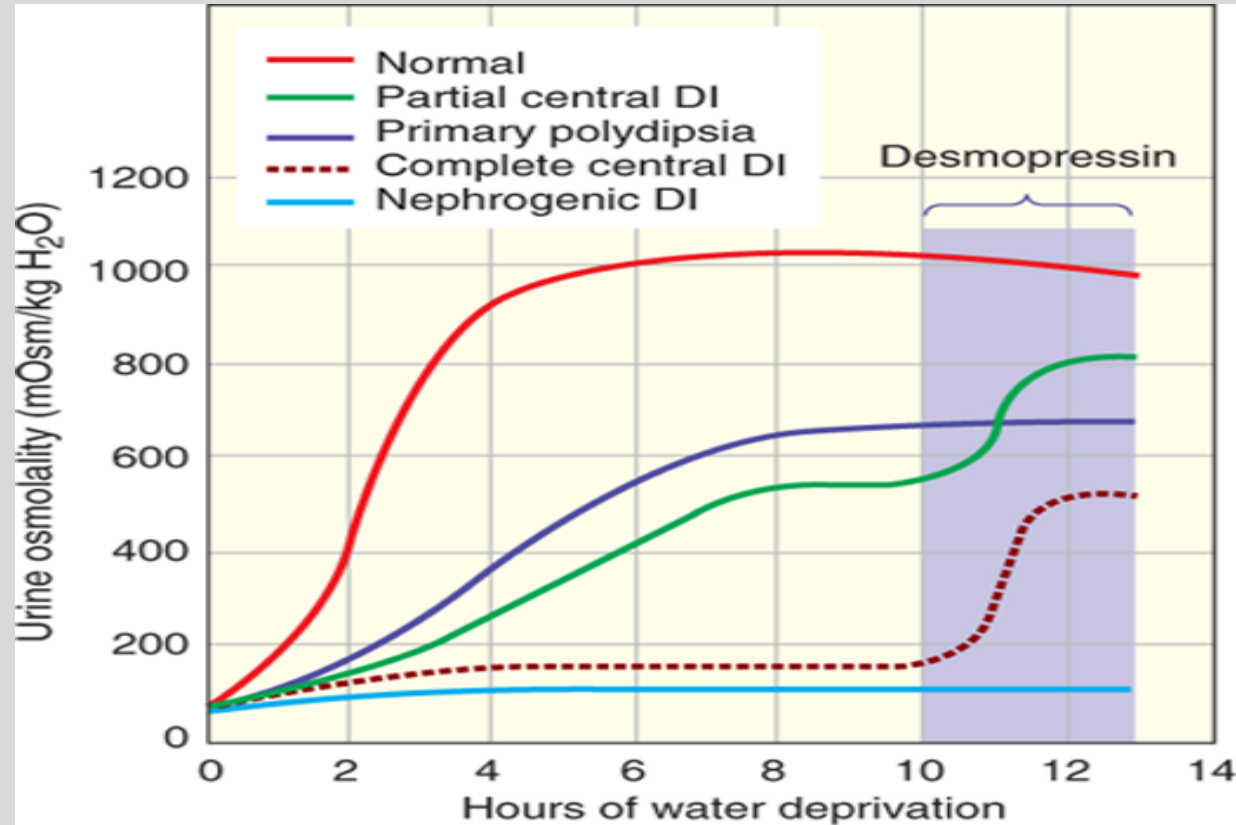


Bichet, D. (2022) Diagnosis of polyuria and diabetes insipidus. *UpToDate*.

Response to Desmopressin

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Source: David G. Gardner, Dolores Shoback: Greenspan's
Basic & Clinical Endocrinology, Tenth Edition
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Treatment

- Central Diabetes Insipidus-
 - Desmopressin
- Nephrogenic Diabetes Insipidus-
 - If possible discontinue offending agent/ correct underlying disorder (Lithium-induced nephrogenic DI may be irreversible if tubular injury is severe)
 - Low sodium-low protein diet as tolerated
 - Thiazide diuretic (Hydrochlorothiazide)
 - Amiloride for patients taking Lithium

- **A 32-year-old male with hx HTN presents to clinic in August with irritability and muscle weakness for the last 3 weeks. He works in construction and is constantly thirsty, which he attributes to the heat and humidity. He is drinking more than 2 gallons of water daily, and his coworkers have complained that he is "always in the john."**

- **He is currently on no medications and denies tobacco, alcohol, and illicit use. He denies fevers, nausea, and diarrhea. You send a complete blood count (CBC), comprehensive metabolic panel (CMP), and creatine kinase (CK) that are only significant for a sodium (Na) of 148 mEq/L.**

- **What is your differential diagnosis for this patient's hypernatremia?**
- **What further diagnostic testing is appropriate?**
- **What would the diagnostic testing show?**

Question

23-year-old male presents to the clinic with complaints of polydipsia and frequent, large-volume urination. He has a past medical history of a posterior pituitary tumor. Physical examination reveals dry mucous membranes and decreased skin turgor. Laboratory testing does not demonstrate any evidence of diabetes; however, a urine osmolality of 120 mOsm/kg is measured (normal 500-800 mOsm/kg). Which of the following is the most likely finding on a desmopressin test in this patient?

- A. urine osmolality to 60 mOsm/L
- B. urine osmolality to 110 mOsm/L
- C. urine osmolality to 230 mOsm/L
- D. urine osmolality to 600 mOsm/L
- E. No change in urine osmolality

Question

A 69-year-old female presents to the ED with confusion. She lives alone and has no caretaker. She is disoriented and lethargic. PE unremarkable except for decreased skin turgor and dry mucous membranes. Na 165 meq/L (normal 135 and 145), BUN 31 mg/dL (normal 7 to 20), Cr 1.1 mg/dL (normal .6 to 1.1). Urine osmolality is 1000 mmol/kg (normal 500-800). Serum osmolality is 300 (normal 275 to 290). Which is the most likely cause of this patient's presentation?

- A. Central diabetes insipidus
- B. Nephrogenic diabetes insipidus
- C. Diuretic overuse
- D. Lack of free water intake
- E. Hypoaldosteronism

Summary Slide

- Explain water balance within the body.
 - Illustrate the pathophysiology of hypernatremia.
 - Differentiate the causes of hypernatremia.
 - Illustrate how to diagnose and treat hypernatremia.
 - Explain diabetes insipidus.
-
- Core References:
 - Mount DB. Fluid and Electrolyte Disturbances. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. Harrison's Principles of Internal Medicine, 22nd Edition. McGraw Hill; 2026
 - Stern SC. Hypernatremia Secondary to Inadequate Water Intake. In: Stern SC, Cifu AS, Altkorn D. eds. Symptom to Diagnosis: An Evidence-Based Guide, 4e. McGraw-Hill Education; 2020
 - Stern SC. Differential Diagnosis of Hypernatremia. In: Stern SC, Cifu AS, Altkorn D. eds. Symptom to Diagnosis: An Evidence-Based Guide, 4e. McGraw-Hill Education; 2020.

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